# ENEX12002

## LECTURE NOTES

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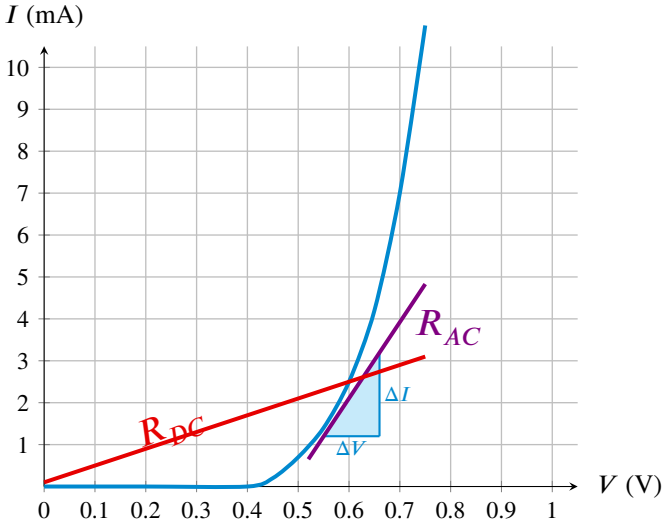
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# 1 Analog Circuits

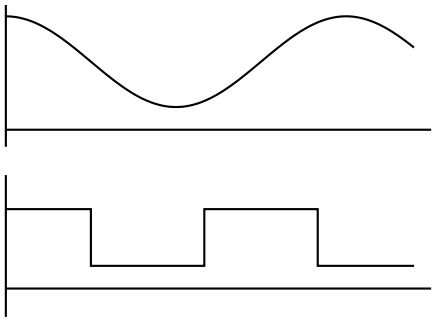
## 1.1 Characteristic Curves:

A characteristic curve is a type of linear plot that demonstates a relationship between two variable properties of a device. Most electronic devices CC refers too a plot of current as a function of coltage. Mostly known as an I-V curve.




## 1.2 Analog vs Digital Signals:

Signalling within a circuit is generally classified as either continuous or discrete. Continuous (Analog signalling) varies after a range of values within limits. Discrete (Digital signals) change between two discreet voltage levels.

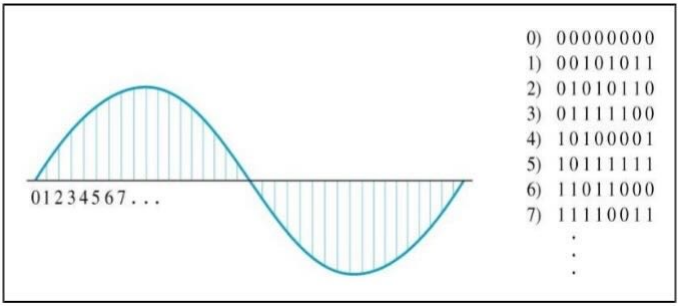


## 1.3 Analog & Digital Circuits:

Analog circuits are used in situations implementing linear functions, wave shaping and transformation between current and voltage. Digital circuit however ase used to process data rapidly and in calculations. Most systems utilize a mix of both systems (e.g AV systems).

## 1.4 Sampling:

When converting between signal types sampling must take place. Sampling is the approximation of analog's original waveform into time slices. The disadvantage of sampling is that is loses some precision, however the advantages of a digital system out weight this. The physical process of sampling is called quantizing. This produces numbers that can be processed by digital systems.




## 1.5 Vectors:

A vector is a line that contains both direction and magnitude. An AC circuits sinusodial waveform can be generated by plotting the project of a rotating vector in circular motion.

## 1.6 Frequency and Period:

The frequency of a periodic wave is the number of cycles that occur every second. The SI unit for this is Hz. The period is how long it takes a waveform to complete one cycle.

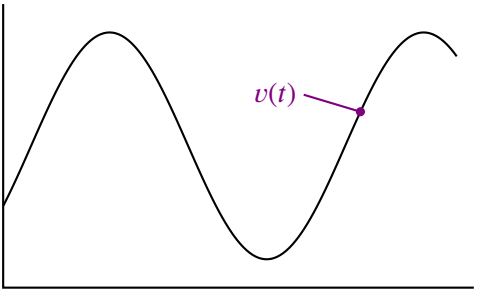
$$f = \frac{1}{T}$$

$$T = \frac{1}{f}$$

## 1.7 Instantaneous Values:

$$v(t) = V_p \sin(\omega t \pm \phi)$$

The instantaneous values are always against time ($t$).




## 1.8 Waveform Values:

The amplitude is the maximum distance of the values from the line of equilibrium. The average value of a sine wave is the found in $|v(t)|$, then averaging the curve.

$$V_{ave} = \frac{2V_p}{\pi} \approx 0.637V_p$$

## 1.9 RMS Values:

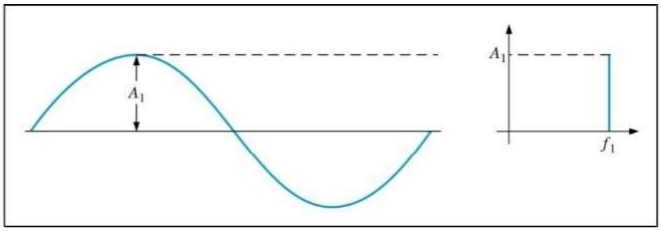
In order to compare AC and DC currents and voltages, AC is converted into its DC equivalent DC heating value. RMS current and voltage are their peak values.

$$\begin{aligned} V_{RMS} =& \frac{V_p}{\sqrt{2}} \\ \approx& 0.707V_p \end{aligned}$$

$$\begin{aligned} I_{RMS} =& \frac{I_p}{\sqrt{2}} \\ \approx& 0.707I_p \end{aligned}$$

## 1.10 Domain Signals:

Time domain is the funcation against time. An example is a osccilisope which shows values over a period of time. The frequency domain - A sine-wave against time can be represented against its frequency. This representations shows the waveforms amplitude and frequency but not it's phase angle.




## 1.11 Transducers:

These are a device that converts a physical quantity (e.g pressure, position or temp) from one form to another. Their outupts are quite small and require amplification. Passice transducers require external power to functions, this is called excitation. Active transducers are self-generating that convert a portion being measured into electrical signal.

## 1.12 Amplification:

Amps are designde to amplify either voltage or power. Linear amplifiers produce a magnified replica of the inputted signal, producing a useful output.

$$A_v = \frac{V_{out}}{V_{in}}$$

## 1.13 Transfer Curves:

All amplifiers, when ideal, produce a linear transfer curves. Beyond this limit they no longer act idealy, this is called saturation.

## 1.14 Power Gain:

$$A_p = \frac{P_{out}}{P_{in}} = \frac{V_{out}I_{out}}{V_{in}I_{in}}$$

### 1.15 dB Power Ratios:

The decibal of a power ratio is:

$$\begin{aligned} A_p &= 10\log_{10}(\frac{P_1}{P_2}) \\ &= 10\log_{10}(A_p) \end{aligned}$$

### 1.16 dB Voltage Ratio:

$$\begin{aligned} A_v &= 20\log_{10}(\frac{V_1}{V_2}) \\ &= 20\log_{10}(A_v) \end{aligned}$$

## 2 Basic Circuit Theory

### 2.1 Electrical Charge:

Electrical effects are attributed to the seperation of charges and their movements. Electrical forces are that opposites attract and similar repels. A charge is considered continuous if $q = 1.6022E\text{-}19e^{-}$.

### 2.2 Voltage and Current:

Rather than working purely in charge, it is easier to work with it in voltage and current.

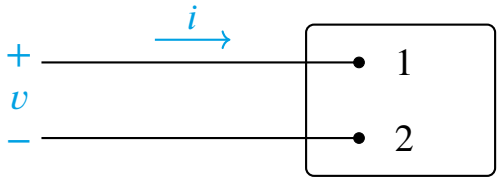
$$V = \frac{dw}{dq}$$

$$I = \frac{dq}{dt}$$

### 2.3 Ideal Basic Circuits:

These are made up of 5 main points:

1. Two Terminals
2. Can be described mathmatically in terms of voltage and current
3. Cannot be subdivided into further elements
4. Utilizes the passive sign convention

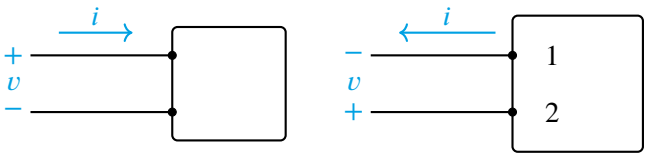



## 2.4 Power and Energy:

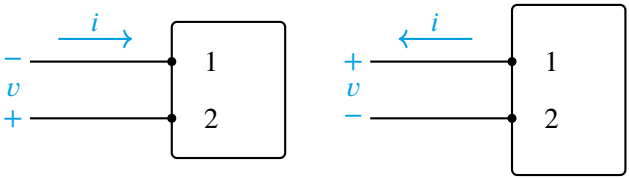
The useful output of a system is often non-electrical. Practical applications also have limitations in the amount of power it can handle.

$$p = \frac{dw}{dt} = (\frac{dw}{dq})(\frac{dq}{dt}) = VI$$

The polarity describes where current flows in the relationship to voltage. If $p > 0$, power is being delivered to the circuit inside the box. For instance a motor.




Else if $p < 0$, power is being extracted from a circuit. For example a battery.



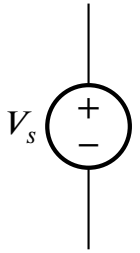

Elements that make up a basic circui include:

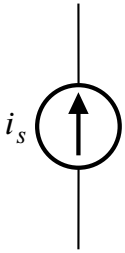
- Voltage source/s
- Current source/s
- Passive Circuit elements, As they do not produce electricity
  - Resistors
  - Inductors
  - Capacitors

## 2.5 Voltage and Current Sources:

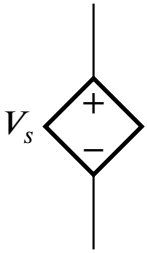
An electrical source that can convert between electrical and non-electrical energy. An ideal voltage source provides a consistant voltage regardless of current. This is similar for current sources. In real-world applications, ideal power sources do not exist.

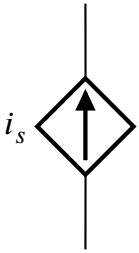
### 2.5.1 Source Types




**Independent Sources**




**Dependent Sources**

## 2.6 Kirchoffs Laws:

A circuit is said to be solved once a voltage and current is found across every circuit element.

$$\text{KCL} \Rightarrow \sum I_{out} = \sum I_{in}$$

$$\text{KVL} \Rightarrow \sum V_{loop} = 0$$

To apply these laws, polarity must be applied throughout the circuit.

## 2.7 Resistance:

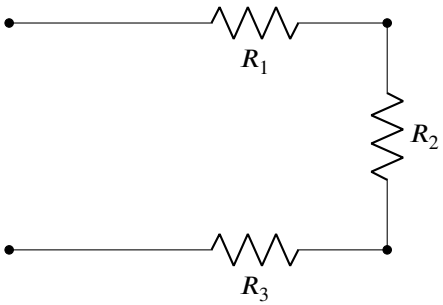
Resistors resist the flow of current within a circuit. Conductance is also the inverse of resistance, $G = \frac{1}{R}$.




$$V = IR$$

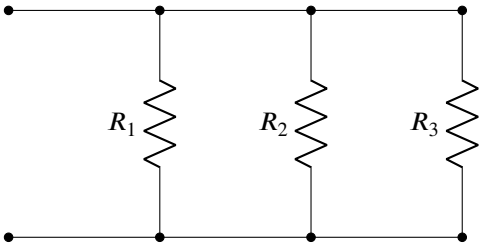
### 2.7.1 Series

$$R_{eq} = \sum_{n=1}^{k} R_n$$



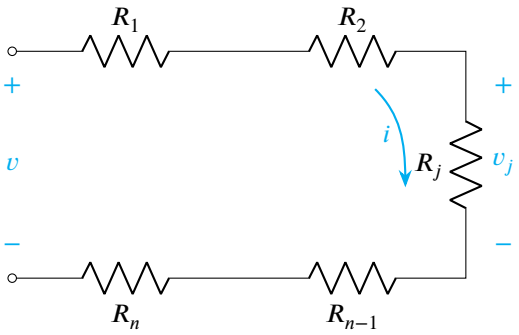

### 2.7.2 Parllel

$$R_{eq} = \frac{1}{\sum^{k} n = 1 \frac{1}{R_n}}$$



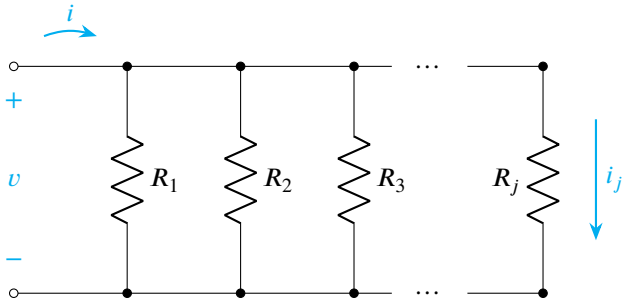

## 2.8 Voltage Division:

$$V_j = iR_j = \frac{R_j}{R_{eq}} V$$




## 2.9 Current Division:

$$i_j = \frac{V}{R_j} = \frac{R_{eq}}{R_j} i$$




## 2.10 Source Transformations:

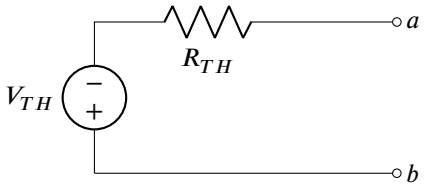
Current and voltage sources can be interchanged. This is because equivalent sources produce the same electrical effects, regardless of load.

### 2.10.1 Thévenin Equivalent

A way to simplify a circuit that focuses on the behaviour at the terminals.

$$V_{TH} = V_{O/C} = V_{ab}$$

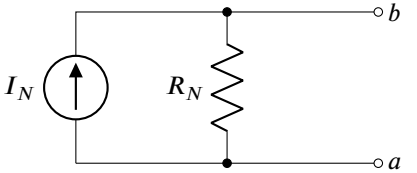
$$R_{TH} = \frac{V_{TH}}{I_{SC}}$$




### 2.10.2 Norton Equivalent

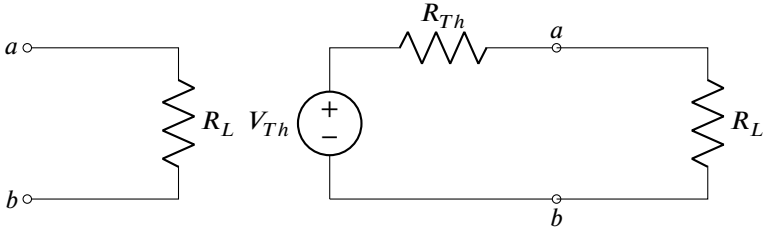
$$I_N = I_{S/C}$$

$$R_N = R_{TH}$$




## 2.11 Maximum Power:

To find a circuits maximum power, replace its resistive network within its Thévenin equivalent.




Power delivery to $R_L$:

$$p = I^2 R_L = (\frac{V_{TH}}{R_{TH} + R_L})^2 R_L$$

Max power is when $R_L = R_{TH}$:

$$p_{max} = \frac{V_{TH}^2 R_L}{(2R_L)^2} = \frac{V_{TH}^2}{4R_L}$$

## 2.12 Superposition Principle:

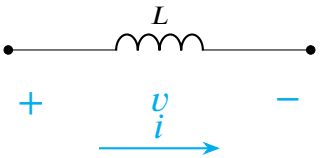
A linear system obeys this principle. Which is when two or more independent energy sources are connected in a circuit, the total response is the sum of each individual.

## 2.13 Inductors:

The behaviour of inductors are based around the properties of electro--magnetism. The SI unit for an inductor is Henry.

$$V = L\frac{di}{dt}$$

$$W = \frac{1}{2}LI^2$$




$$i(t) = \frac{1}{L}\int_{t_0}^{t} v\, d\tau + i(t_0)$$

$$p = Li\frac{di}{dt} = \frac{dw}{dt} = vi(t)$$

### 2.13.1 Series

$$L_{eq} = \sum_{n=1}^{k} L_n$$

### 2.13.2 Parallel

$$L_{eq} = \frac{1}{\sum_{n=1}^{k} \frac{1}{L_n}}$$

## 2.14 Capacitor:

Made-up of two conductors seperated by an insulator. It acts as an O/C to DC current and S/C to AC. An important note to remember that electrical charge is not directly transported through the capacitor.




$$V = IR$$

$$i = C\frac{dv}{dt}$$

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i\, d\tau + v(t_0)$$

$$p = vi = Cv\frac{dv}{dt}$$

$$w = \frac{1}{2}Cv^2$$

### 2.14.1 Series

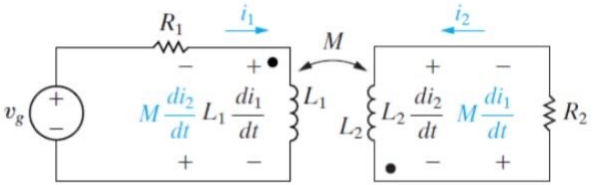
$$C_{eq} = \frac{1}{\sum_{n=1}^{k}\frac{1}{C_n}}$$

### 2.14.2 Parallel

$$C_{eq} = \sum_{n=1}^{k} C_n$$

## 2.15 Mutual Inductance:

The magnetic fields between the two circuits create a mutual and self-
-inductance voltages. Polarity of M depends on the way the coils are wound. $L_1$ and $L_2$ must be wound is opposing directions.



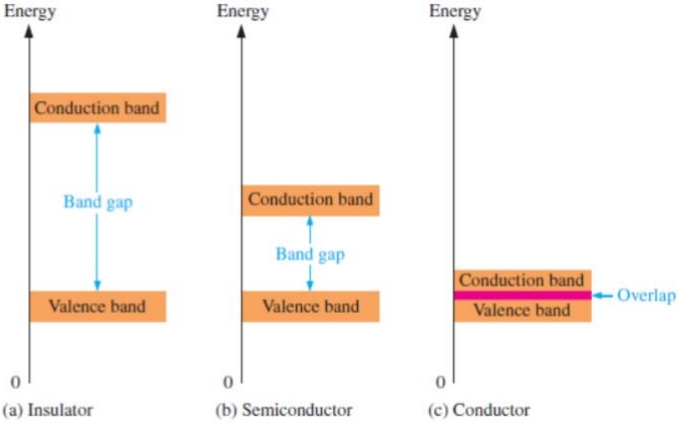

# 3 Diodes

## 3.1 Chemistry Revision:

Electrons form themselves into shells, with the outer shell containing the most energy due to their great mobility. These electrons in the outer shell are called valence electrons. Ions are formed through either an element donating or recieving electrons (e.g. Cations and Anions)

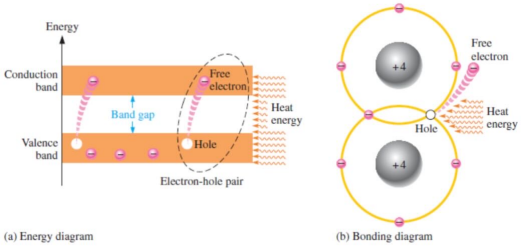
## 3.2 Energy Gaps:

Valence bonds within inter-elemental bonds are in the valence band. A conduction band is the energy band beyond the valence band.




| Material | $\rho(\Omega \cdot cm)$ |
|---|---|
| Insulator | $\rho > 10^5$ |
| Semiconductor | $10^{-5} < \rho < 10^5$ |
| Conductor | $\rho < 10^{-5}$ |

## 3.3 Electrons and Holes:




In instrinsic silicon it's current is produced by the movement of free electrons toward the end. The effective movement of the hole is opposite direction of electron movement.

## 3.4 Silicon:

### 3.4.1 Doping Silicon

#### 3.4.1.1 Donor Atoms

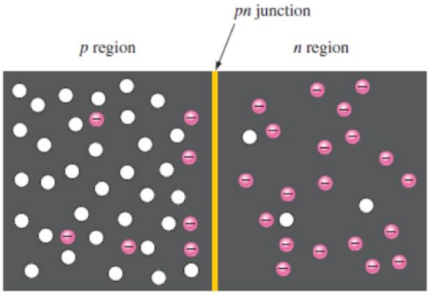
These atoms are pentavalent imparity, with 5 electrons in the valence shell. They are doped into the silicon crystal structure that donates the extra electron. This then becomes a conduction electron.

#### 3.4.1.2 Acceptor Atoms

Place a trivalent imparity into the silicon crystal structure. Creating a hole within the silicon molecule.

### 3.4.2 PN Junction

In the image below, the acceptor is the *p*-region and donor is *n*-region. Within the PN-junction, the *p*-region is negatively charged, and the *n*-region is slightly positively charge.




The PN-junction forms a basic diode. This is something that only allows current flow in one direction. A voltage drop is formed across the reigon as depletion of *n*-type silicon and equillibrium occurs when it repel electron migration.

## 3.5 Diodes:

### 3.5.1 Forward Bias

An external voltage assist electrons in moving through the PN-junction, allowing higher current flow. This reduces the width of the PN-junction.

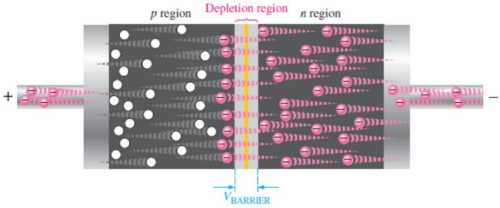
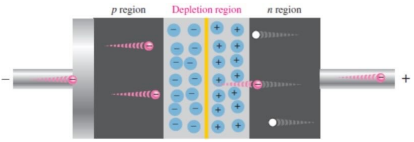



FIGURE 2–4
A forward-biased diode showing the flow of majority carriers and the voltage due to the barrier potential across the depletion region.

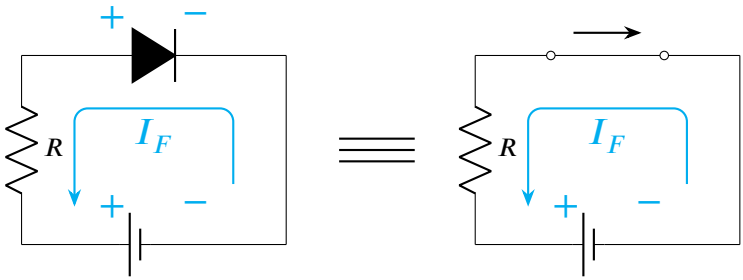
### 3.5.2 Reverse Bias

An extremely small reverse current pushes electrons back towards the PN--junction.



## 3.6 Ideal Diode Models:

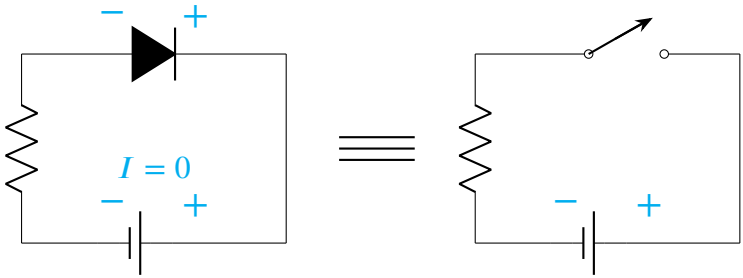
### 3.6.1 Forward Bias




$$I_F = \frac{V_{BIAS}}{R_{LIMIT}}$$

$V_F = 0V$, acts as S/C

### 3.6.2 Reverse Bias

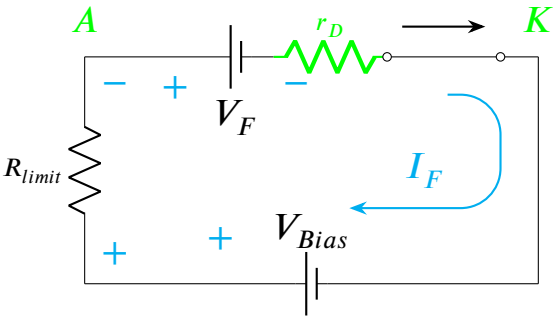



$$I_R = 0$$

$V_R = V_{BIAS}$, acts as O/C

## 3.7 Practical Diode Models:

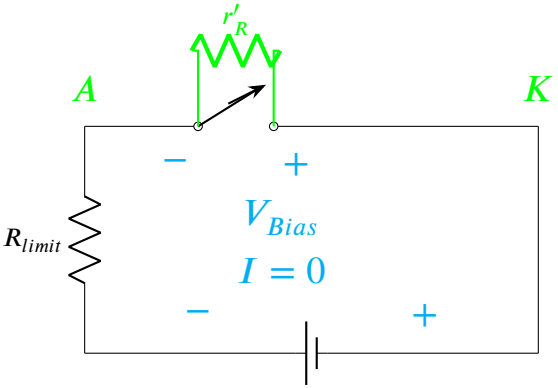
### 3.7.1 Forward Bias




$$V_F = 0.7 + I_F r'_d$$

$$I_F = \frac{V_{BIAS} - 0.7}{R_{LIMIT} + r'_d}$$

### 3.7.2 Reverse Bias

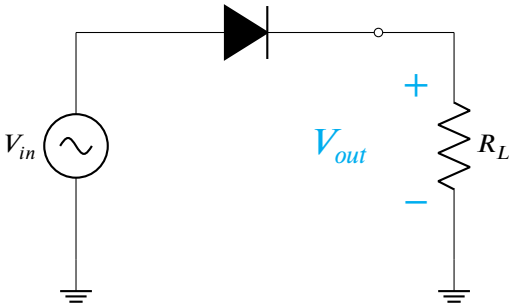



The cathode side of the diode is connected to positive end of the power supply. Whereas, the anode is connected to the negative side.

# 4 Diode Applications

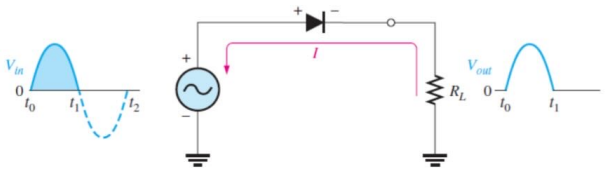
## 4.1 Wave Rectifiers:

A rectifieris a type of circuit configurations that converts AC to its pulsating DC equivalent. A half-wave rectifier only converts half of the AC sine wave (e.g. positive or negative). A full wave rectifier converts the entirety of the sine wave.

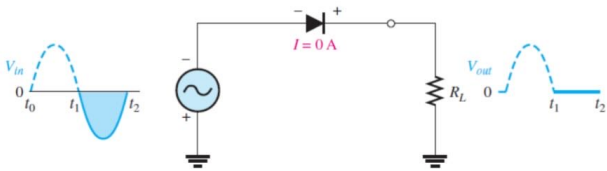



For the circuit above, if $V_{in}$ is negative or smaller than the voltage barries in the diode, the diode should be reversed bias.

### 4.1.1 Half-wave operation




(a) During the positive alternation of the 60 Hz input voltage, the output voltage looks like the positive half of the input voltage. The current path is through ground back to the source.



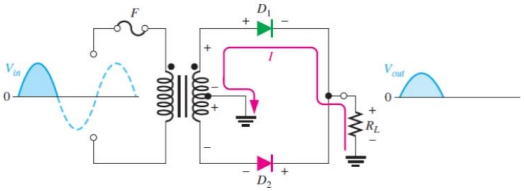

(b) During the negative alternation of the input voltage, the current is 0, so the output voltage is also 0.



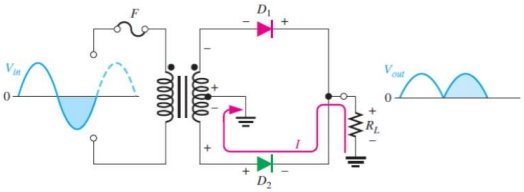

(c) 60 Hz half-wave output voltage for three input cycles

### 4.1.2 Full-wave Rectifier

This circuit controls the direction of the load currents so that only the positive or negative outputs are recieved.

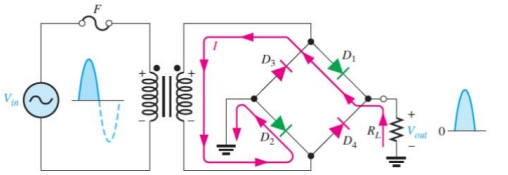



(a) During positive half-cycles, $D_1$ is forward-biased and $D_2$ is reverse-biased.

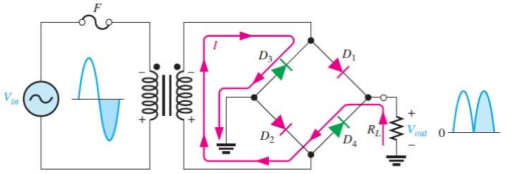



(b) During negative half-cycles, $D_2$ is forward-biased and $D_1$ is reverse-biased.

### 4.1.3 Bridge Rectifier




(a) During the positive half-cycle of the input, $D_1$ and $D_2$ are forward-biased and conduct current. $D_3$ and $D_4$ are reverse-biased.

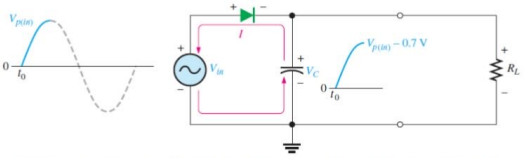



(b) During the negative half-cycle of the input, $D_3$ and $D_4$ are forward-biased and conduct current. $D_1$ and $D_2$ are reverse-biased.

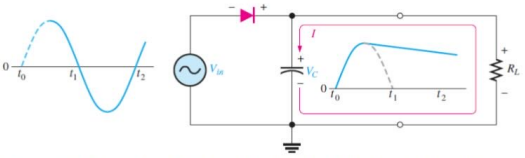
### 4.1.4 Smoothing Rectified Signal

The ripple within the output signal can put strain on connected DC components. A capcitor is used to smooth the signal when the voltage no longer charges the

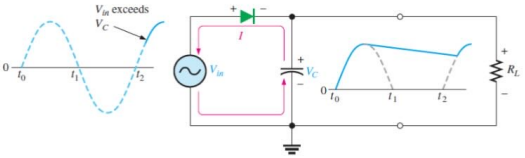
capacitor. The larger the capacitor the small the ripple, and smoother the output voltage will be.




(a) Initial charging of the capacitor (diode is forward-biased) happens only once when power is turned on.




(b) The capacitor discharges through $R_L$ after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid dark blue curve.



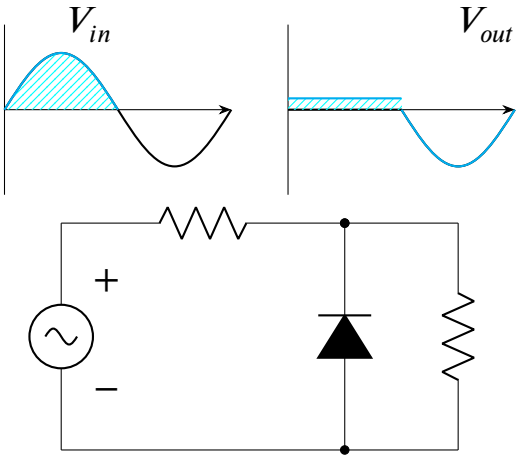

(c) The capacitor charges back to peak of input when the diode becomes forward-biased. This charging occurs during the portion of the input voltage indicated by the solid dark blue curve.

## 4.2 IC Regulators:

An intergrated circuit (IC) is a complete circuit on a small chip of silicon. The IC regulator is an IC connected to the output of a rectifier maintaining a consistent output voltage. The most common ones are 3-pin. The pins are for input, output, and reference adjustment.

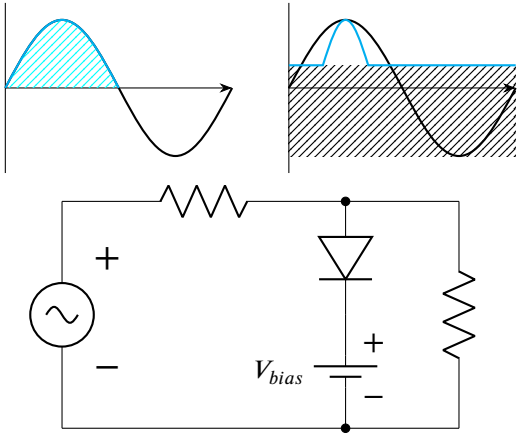
## 4.3 Diode Limiters:

Limits the voltage to a certain value within the diode itself. The voltage is only limited to the circuit parallel to the diode. A common example of this is the Zener diode.

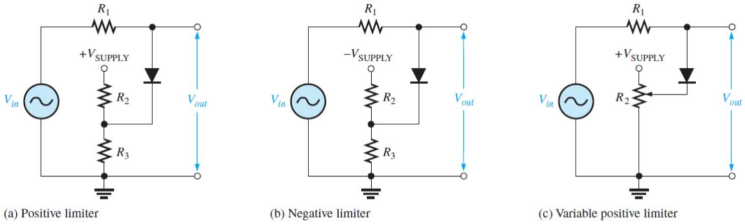



## 4.4 Bias Limiters:

Similar to a diode limiter, but maintains a stable minimum voltage.



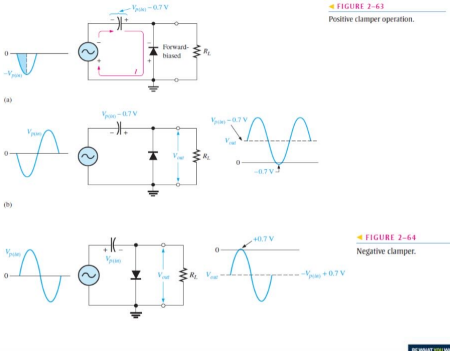
## 4.5 Bias Variation:




(a) Positive limiter (b) Negative limiter (c) Variable positive limiter

## 4.6 Diode Clampers:

Adds a DC offset to an AC waveform. These can also be known as DC restorers.




## 4.7 Extras:

PIV of a diode is just the complete negative of the input voltage. As the diode acts like a S/C. All diodes in series drop 0.7V per diode.
$V_{RMS}$ of half wave rectifier is:

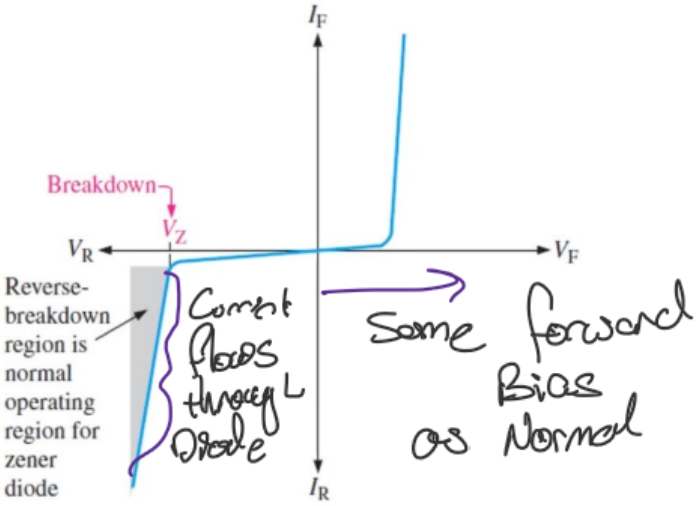
$$V_{RMS} = \frac{1}{\sqrt{2}} \cdot \frac{V_{out}}{2}$$

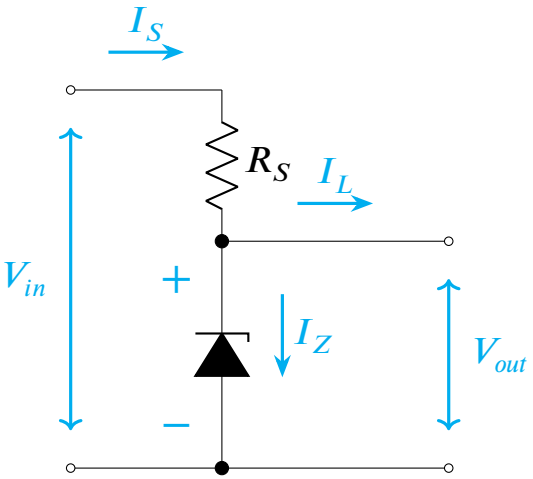
$$V_{avg} = \frac{4V_{out}}{\pi}$$

# 5 Special Diodes

## 5.1 Zener Diode:

A Zener diode is one which mainly operates in its reverse break-down region.



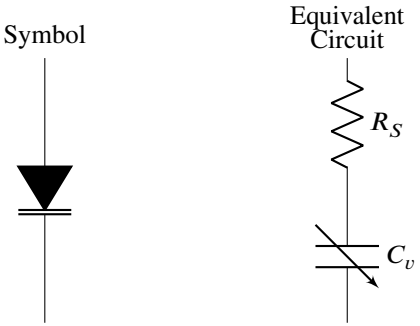

$$R_S = \frac{V_{in} - V_{out}}{I_Z}$$

$$I_{Z(load)} = I_{Z(noload)} - I_L$$

The breakdown voltage, is the ideal values for voltage items. After the maximum current through the diode. If more current is applied to the diode, permenant damage occurs. The knee current is the minimum current required for the voltage regulation to work properly.

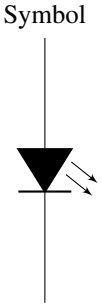
## 5.2 Varactor Diode:

This diode is a reversed-biased PN-junction that utilizeds the inherent capacitence of the depletion region.




## 5.3 LightEmtting Diodes:

LEDsemit light when exposed to power. As such they are very useful as indicator throughoug a circuit. These diode types are designed to work only in forward bias conditions.




### 5.3.1 Organic LED:

The way OLED works is identical to the normal LEDS. The make-up of an OLED display is 2-3 layers, composed of organic polymers that emit light on

certain voltage levels.

### 5.4 Photodiode:

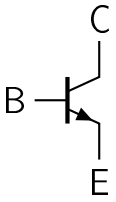
A ligh absorbing device that conducts current based on light levels. This works in reverse bias configuration.

Symbol

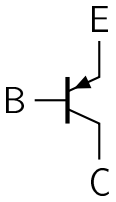


## 6 Bipolar Junction Transistor

The following symbols are used to denote between the two types present. In where the types of silicon are placed within the silicon.



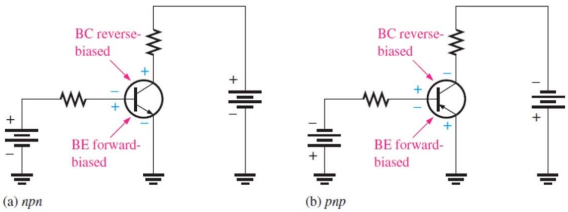

npn




pnp

### 6.1 BJT Biasing:

For a BJT to operate properly, it must be biased. The collector must be in reverse bias and emitter forward. The current direction opposes the direction the arrow is pointing.




$$I_E = I_B + I_C$$

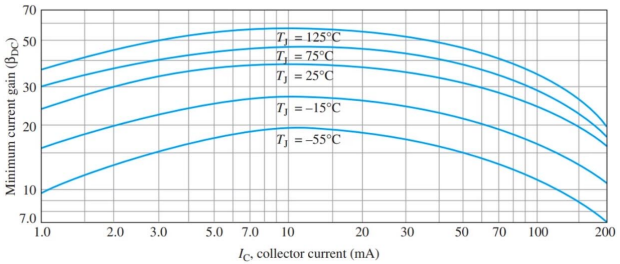
Since base current is much lower than emittor or collector current the can equation equates to roughly $I_E \approx I_C$.

## 6.2 Beta DC:

$\beta_{DC}$ is the current gain of a transistor. On a datashett $\beta_{DC}$ is usually $h_{FE}$. This is only valid if the transistor operates within the manufacters specified range.

$$\beta_{DC} = \frac{I_C}{I_B}$$

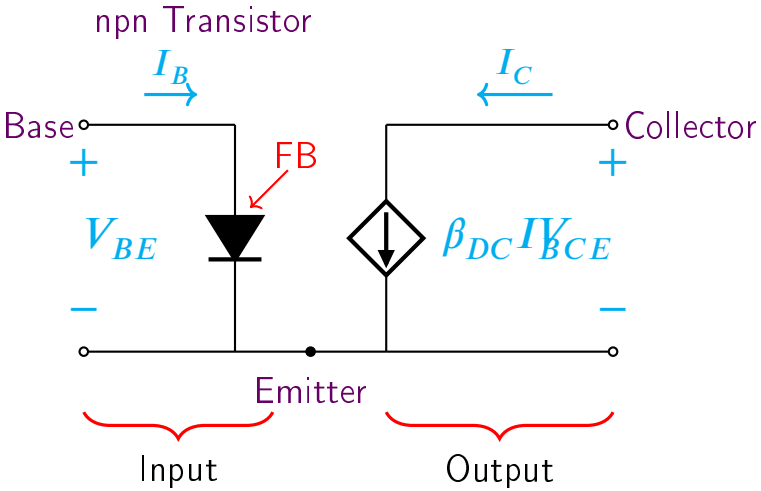
$\beta_{DC}$ is not always a consistent value, as shown in the image below.



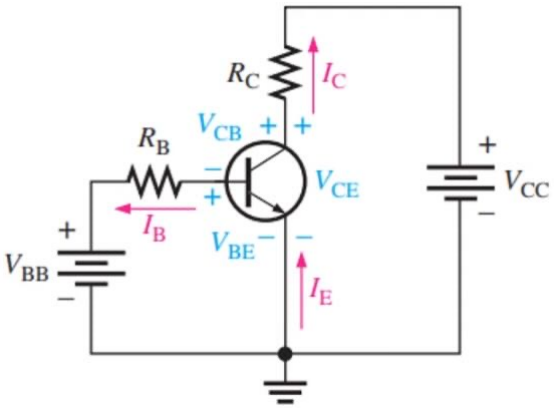

## 6.3 Alpha DC:

$\alpha_{DC}$ is the ratio of the DC collector to DC emittor currents. Typical value ranges for $\alpha_{DC}$ is $0.95 \rightarrow 0.99$. It is less common for $\alpha_{DC}$ to be found in the components datasheets than $\beta_{DC}$.

## 6.4 Equivalent Circuits:




## 6.5 Voltage and Current:



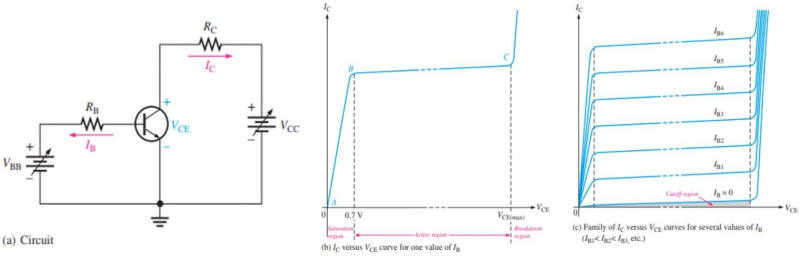

$$V_{BE} = 0.7V$$

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CB} = V_{CE} - V_{BE}$$

## 6.6 Collector Characteristic Curves:




## 6.7 Cut-off:

When $I_B \approx 0$, the BJT is cut off and there is no collector current. In cut-off both base-emitter and base-collector are in reverse bias.

## 6.8 Saturation:

When base current is large, the $V_{CC}$ drop across $R_C$ is sufficient, leaving no voltage between the collector and emitter.

$$I_{C(sat)} = \frac{V_{CC}}{R_C}$$

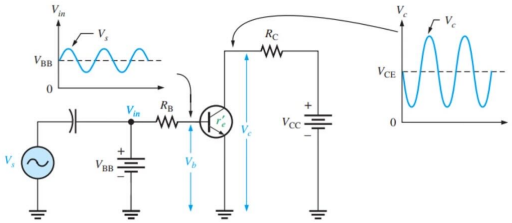
$$I_{C(sat)} = \frac{V_{CC} - V_{CE(sat)}}{R_C}$$

## 6.9 Maximum Rating:

Transistors have limitations specified on the datasheet. As such the $V_{CE}$ and $I_C$ of the component must not exceed the $P_{D(max)}$. At higher temps the $P_{D(max)}$ is less.

## 6.10 BJT as an Amplifier:

$$V_{out} = \frac{R_C}{r'_e} \times V_b$$




## 6.11 BJT as a Switch:

By switching a BJT in either cut-off or saturation, a basic switch can be formed. This is either through letting current not flow or flow through the transistor.

